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Effects of risk-based pricing reform on flood insurance uptake

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Abstract

Insurance is critical for financial recovery following climate-related disasters, particularly for low-income households. As premiums increase in response to the growing cost of climate impacts, there are concerns that coverage may become unaffordable for many households, leading to a decline in insurance uptake. In 2021 and 2022, the National Flood Insurance Program (NFIP) began a major pricing reform, known as Risk Rating 2.0, to better align premiums with property-level flood risks and to improve the solvency of the program. We aim to understand the impacts of this reform on flood insurance demand by estimating the effects of NFIP premium changes on the number of policies-in-force. Using difference-in-differences models with dynamic treatment effects, we find that Risk Rating 2.0 has caused an 11 – 39% decline in new policies and a 5 – 13% decline in existing policies, depending on the amount premiums have increased. These effects are larger in zip codes with lower median household income, whereas differences across flood zones are not statistically significant. Despite the potential benefits of risk-based pricing reform, our results highlight the need for policy interventions that help low-income households maintain their insurance coverage.

Key Points

1. A recent reform to the National Flood Insurance Program's pricing methodology, known as Risk Rating 2.0, has caused substantial premium increases for many policyholders.
2. Risk Rating 2.0 has caused a 11 – 39% decline in new policies and a 5 – 13% decline in existing policies, depending on the amount premiums have increased.
3. The effects of Risk Rating 2.0 on flood insurance uptake are relatively larger in zip codes with lower median household income.

Introduction

Insurance is critical for managing financial risks and supporting recovery following climate-related disasters (Collier and Kousky, 2024; Kousky, 2019; You and Kousky, 2024). As the costs associated with climate impacts continue to grow, insurers will need to increase rates commensurate with the growing risk of loss in order to maintain profitability (Wagner, 2022b). This is also true for public insurance programs needing to remain solvent. In the United States, for example, expected annual losses to residential properties from flooding currently exceed \$32 billion and are predicted to increase by at least 26% by 2050 (Wing *et al.*, 2022). Reflecting this risk in insurance rates will be necessary for fiscal soundness and to align incentives for risk reduction. However, higher premiums may decrease insurance demand, leaving many households financially vulnerable in the event of flooding and potentially adding to public recovery costs (Deryugina, 2017; You and Kousky, 2024).

Since it was first introduced in the late 1960s, the National Flood Insurance Program (NFIP), housed within the Federal Emergency Management Agency (FEMA), has been the primary means for U.S. homeowners and small businesses to insure against flood damages. As of 2018, the NFIP underwrote 90 to 95% of all residential flood insurance contracts (Kousky *et al.*, 2018). Unlike homeowner's insurance, which is required by nearly all mortgage lenders, flood insurance is only required for households with federally-backed mortgages located within FEMA's Special Flood Hazard Area (SFHA; i.e., areas with a $\geq 1\%$ annual probability of flooding). Despite this requirement, only 48% of households within the SFHA had flood insurance as of 2019; outside of the SFHA, where coverage is voluntary, only 2.2% of households had flood insurance (Bradt *et al.*, 2021).

NFIP premiums have historically been priced below actuarially sound rates, as evidenced by the program's \$20 billion debt to the U.S. Treasury (Congressional Research Service, 2025). This mispricing was due to a combination of factors, including outdated Flood Insurance Rate Maps (FIRMs), the failure to account for pluvial

flood risk and changing climate, and a range of discounts and subsidies for certain properties (Kousky *et al.*, 2017). In addition to the fiscal burden this creates for the federal government (Office of Management and Budget, 2022), underpriced flood insurance premiums have created perverse incentives for excess development in high risk areas and underinvestment in hazard mitigation (Peralta and Scott, 2024; Colby and Zipp, 2021; Fabian, 2024).

To address these issues, the NFIP recently began a major pricing reform, known as Risk Rating 2.0. This reform has aimed to harness improvements in data and modelling to shift premiums towards actuarially-sound rates that more accurately reflect property-specific flood risk. Risk Rating 2.0 took effect for new policies on October 1, 2021, and for renewals of pre-existing policies on April 1, 2022. New policies were immediately subject to the full risk-based rate, whereas premium increases for most existing policies are capped at 18% per year¹. For these existing policies, premiums continue to increase at a rate of 18% per year until they reach their full risk-based rate.

The new pricing methodology moves the NFIP away from its reliance on coarse flood zone designations, particularly the 100-year floodplain, and towards an individual assessment of risk for each property more aligned with industry best-practice. Premiums for each property are now determined based on individual factors that include expected annual flood losses from an ensemble of three privately developed catastrophe models, coupled with property-specific rebuilding costs (Horn, 2021). This new pricing model alters premium costs for almost all policyholders, but does not alter flood maps or change the mandatory purchase requirement for households with federally-backed mortgages located within the SFHA.

While Risk Rating 2.0 has the potential to improve the solvency of the NFIP by aligning premiums with property-level flood risks, these pricing reforms may create trade-offs with affordability and insurance uptake (de Ruig *et al.*, 2022; Mulder and Kousky, 2023). Past experience shows that when insurance prices increase, many homeowners, particularly those with lower

1. The NFIP is not allowed to increase premiums beyond the statutory limits set in the Homeowner Flood Insurance Affordability Act of 2014, which allows premium increases of up to 18% annually for primary residences. However, other categories of properties, such as non-primary residences, non-residential properties, business properties, and properties with severe repetitive loss, are allowed to have premium increases of up to 25% annually.

incomes, simply drop or reduce coverage (Hennighausen *et al.*, 2023; Wagner, 2022a; Sastry *et al.*, 2025; Mulder, 2024), and maintain their location choices (Collier *et al.*, 2023), leaving them financially vulnerable in the event of flooding (You and Kousky, 2024).

We assess these potential unintended consequences of risk-based pricing under Risk Rating 2.0 by estimating the effects of the reform on NFIP policy uptake in the months since its implementation. This analysis builds on a similar study on Risk Rating 2.0 by Ortega and Petkov (2025) in three ways: 1) we estimate dynamic treatment effects to understand changes in policy uptake over time; 2) we estimate heterogeneous treatment effects by household income to understand the distributional impacts of Risk Rating 2.0; and 3) we estimate unique treatment effects for three ordinal categories of expected premium increases under Risk Rating 2.0. We find that Risk Rating 2.0 has caused an 11 – 39% decline in new policies and a 5 – 13% decline in existing policies, depending on the amount premiums have increased, as compared to areas with the smallest expected price changes. These effects are larger in zip codes with lower median household income, whereas the differences in effects between SFHA and non-SFHA policies are not statistically significant.

Data & Methods

In April 2023, FEMA published summaries of NFIP premium changes for single-family homes under Risk Rating 2.0 at the zip code level². The data release included the median cost of insurance in September 2022 (Figure 1A) and the median risk-based cost of insurance under Risk Rating 2.0 (Figure 1B) for each zip code with more than five policies-in-force (PIF) as of September 2022. The median risk-based costs reflect what existing policyholders (as of 2022) would pay if they were paying the full actuarial rate, based on expected losses and programmatic expenses, without any discounts. Across zip codes, the median change in premiums between the current and full risk-based rates is US\$433 (Figure 1C) or 63% (Figure 1D). The distributions of premium increases are highly skewed and have long right tails. While many existing policyholders experience minimal changes, in the

top deciles, premiums are expected to ultimately increase by US\$2,072 or 279% under the full risk-based rates.

We accessed data on NFIP policy transactions from the OpenFEMA website³. This dataset includes a history of individual flood insurance contracts between 2009 and 2024. Each transaction includes the effective date and termination date of the policy (typically one year duration), the original “new business” date of the policy, as well as the flood zone status, occupancy type, and zip code of the property. Using these data, we determine whether each observation is a “new” policy or a renewal of an “existing” policy. A policy is considered “new” if the original date of the policy (i.e., “new business” date) is the same as its effective date; otherwise, the transaction is considered a renewal. We then count the number of “new” and “existing” residential PIF in each zip code on a rolling monthly basis over time. For example, if a new policy was created in June 2023, then it would be included in the new PIF count for December 2023; then, if the policy is renewed in June 2024, then it would be included in the existing PIF count for the following year.

We estimated the effects of these expected price changes on NFIP policy uptake from the date of implementation (i.e., October 2021 for new policies and April 2022 for existing policies) through October 2024 using difference-in-differences models with dynamic treatment effects (Eq. 1; Eq. 2). Our outcome variables are the number of new PIF (Eq. 1) and existing (Eq. 2) PIF, in zip code i in month-year t . The PIF variables were log transformed to ensure normality, since their distributions are highly right-skewed (Figure 1E; Figure 1F). We separated these outcomes and their associated models because all new policies were immediately subject to the full risk-based rate, whereas premium increases for most existing policies are capped at 18% per year – effectively creating two different treatment regimes with potentially heterogeneous effects on NFIP uptake. This approach necessarily assumes that premiums for new policies in each zip code have the same median risk-based rate as policies that existed as of September 2022, since new policies are not reflected in the summaries of NFIP premium changes published by FEMA.

2. Available at: <https://www.fema.gov/flood-insurance/work-with-nfip/risk-rating/single-family-home>

3. Available at: <https://www.fema.gov/openfema-data-page/fima-nfip-redacted-policies-v2>

$$\log (NewPIF_{it}) = \sum_{k=-12}^{k=36} \beta_k^2 D_k^t Q_i^2 + \sum_{k=-12}^{k=36} \beta_k^3 D_k^t Q_i^3 + \sum_{k=-12}^{k=36} \beta_k^4 D_k^t Q_i^4 + \lambda_i + \gamma_{ct} + \varepsilon_{it} \quad \text{Eq. 1}$$

$$\log (ExistPIF_{it}) = \sum_{k=-12}^{k=30} \beta_k^2 D_k^t Q_i^2 + \sum_{k=-12}^{k=30} \beta_k^3 D_k^t Q_i^3 + \sum_{k=-12}^{k=30} \beta_k^4 D_k^t Q_i^4 + \lambda_i + \gamma_{ct} + \varepsilon_{it} \quad \text{Eq. 2}$$

We established treatment and control groups by splitting our sample of zip codes into four quartiles based on the percent change between the median premium in September 2022 and the median full risk-based rates under Risk Rating 2.0 (see Figure 1D). Premiums in zip codes in the first quartile (Q^1) are expected to increase by 0 – 8%; in the second quartile (Q^2), premiums are expected to increase by 8 – 34%; in the third quartile (Q^3), premiums are expected to increase by 34 – 94%; in the fourth quartile (Q^4), premiums are expected to increase by >94%. These quartile groupings do not directly account for the 18% annual cap on premium increases; they instead represent the percent difference between the pre-Risk Rating 2.0 rates and the full risk-based rates under Risk Rating 2.0. Since there are no zip codes that are “untreated” by Risk Rating 2.0, we use zip codes in the first quartile as a comparative control group. For all other zip codes we use binary treatment variables, Q_i^2 , Q_i^3 and Q_i^4 , equal to one if zip code, i , is in the second, third, or fourth quartile, respectively. Dummy variables, D_k^t , are equal to one when month-year t is equal to event time k . Our coefficients of interest are β_k^2 , β_k^3 , and β_k^4 which estimate average treatment effects on PIF at k months relative to October 2021 in Eq.1 and April 2022 in Eq. 2. These estimated effects for each treatment group are relative to our control group of zip codes in the first quartile. This approach therefore assumes that, in the absence of treatment, each of our three treatment groups would follow the same trends over time as the control group.

We also include unit fixed effects, λ_i , to control for time invariant zip code characteristics and county-by-month-year fixed effects, γ_{ct} , to control for localised changes in NFIP policy uptake over time that may be associated with the occurrence of flood events (e.g. Choi *et al.*, 2024), or other localised economic shocks.

As a robustness check, we also specify our model using a Poisson regression, where the outcome variables are PIF counts and not log transformed.

We explore heterogeneity in our estimated treatment effects according to policies’ flood zone status and the median household income in zip codes. For flood zone status, we subset the number of PIF in the outcome variable depending on whether the policies are located within or outside of the SFHA. For median household income, we subset the estimation sample by income quartiles. This approach implicitly assumes that the median household income across zip codes, based on the American Community Survey sample of households in a zip code, is tightly correlated with the median household income of NFIP policyholders across zip codes. Due to the mandatory purchase requirement, we would expect premium increases to have less of an effect within the SFHA, where households with federally-backed mortgages are statutorily required to carry flood insurance. Conversely, low-income households may be more likely to drop coverage in response to rising premiums as their ability to increase expenditures is more constrained (Rampini and Viswanathan, 2013; Ericson and Sydnor, 2018).

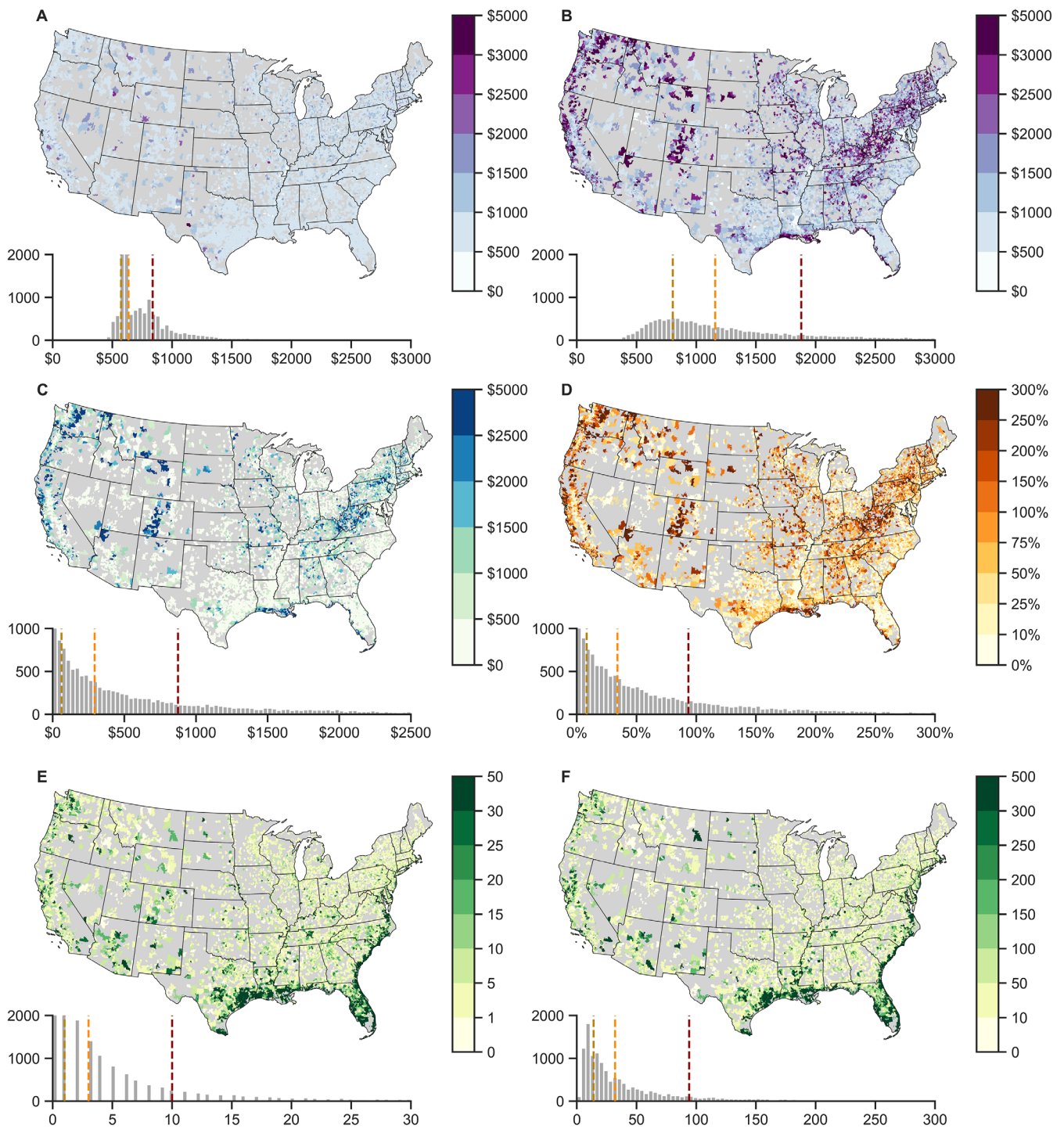


Figure 1: Summary of NFIP pricing changes under Risk Rating 2.0. A) Median premium in zip code as of September 2022. B) Median full risk-based premium in zip code under Risk Rating 2.0 pricing methodology. C) Difference between panels A and B in dollars. D) Difference between panels A and B in percent. E) Number of new PIF in September 2022. F) Number of existing PIF as of September 2022. Vertical dashed lines on histograms indicate 25th, 50th, and 75th percentiles.

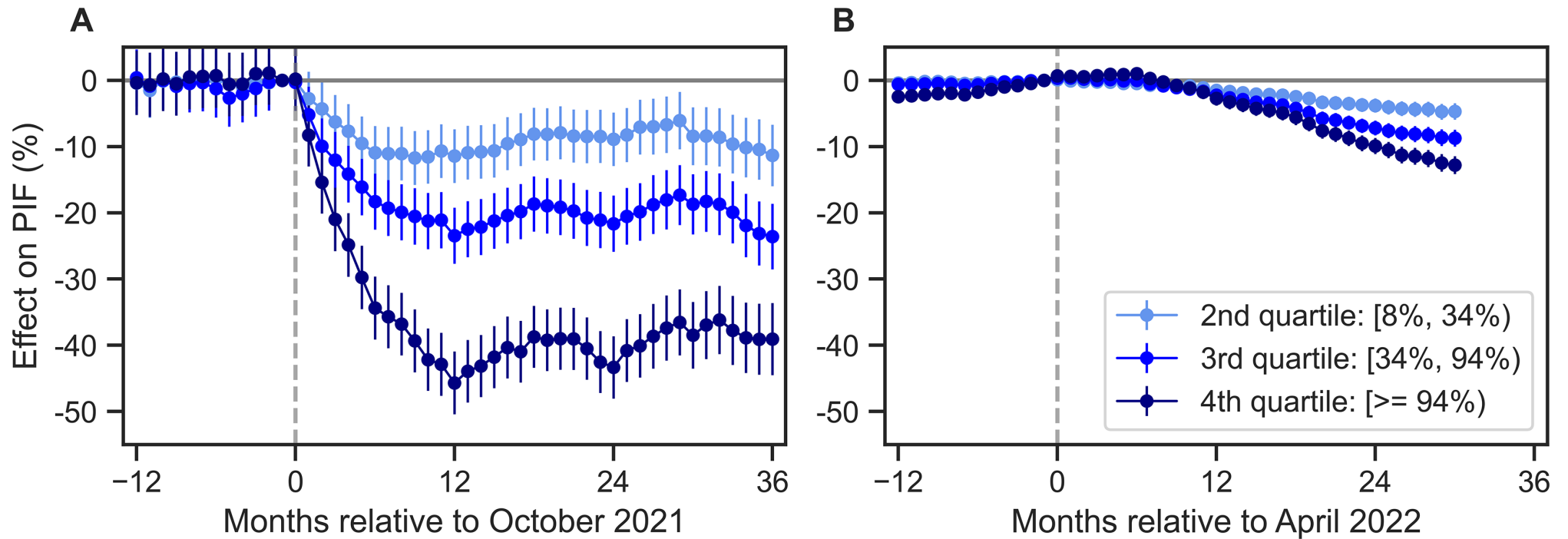


Figure 2: Estimated effects of Risk Rating 2.0 on NFIP policies-in-force by month relative to date of implementation. A) Estimated coefficients for new policies. B) Estimated coefficients for existing policies. Error bars indicate 95% confidence intervals.

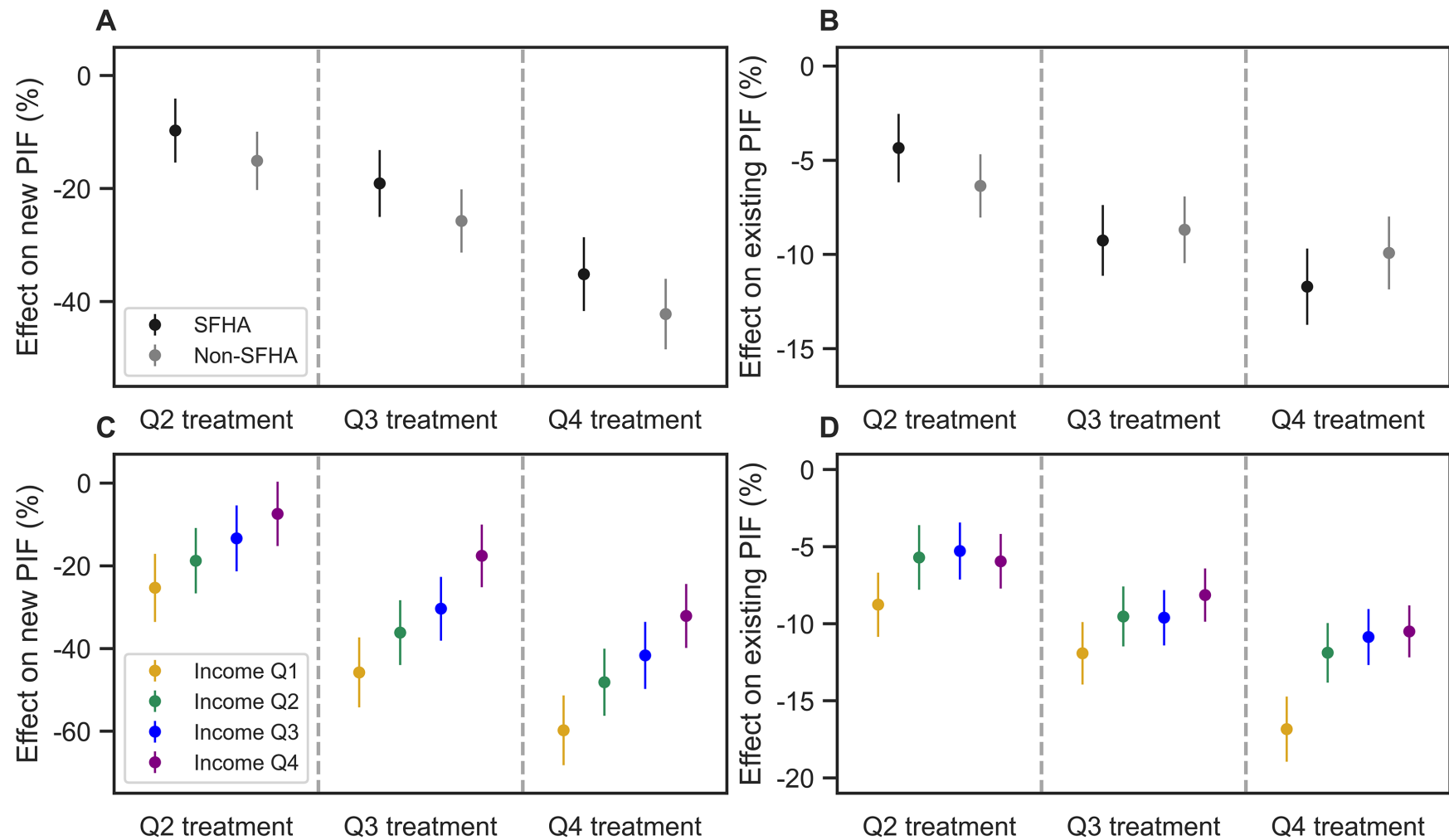


Figure 3: Averaged effects of Risk Rating 2.0 on NFIP policies-in-force in May 2024 through October 2024. A) Estimated coefficients for new policies by flood zone. B) Estimated coefficients for existing policies by flood zone. C) Estimated coefficients for new policies by median household income. D) Estimated coefficients for existing policies by median household income. Error bars indicate 95% confidence intervals.

Results

NFIP premium increases under Risk Rating 2.0 have led to a decrease in policy uptake. Relative to the control group, the number of new policies declined by 11% in the Q2 treatment group, 24% in the Q3 treatment group, and 39% in the Q4 treatment group – as of October 2024 (Figure 2A). These estimates imply demand elasticities of -1.38 to -0.32 for the Q2 treatment group, -0.71 to -0.26 for the Q3 treatment group, and greater than -0.41 for the Q4 treatment group. The decline in new policies began immediately following the implementation of the new pricing methodology and continued for roughly 12 to 24 months (depending on the treatment group). It appears that the decline in new policies may have plateaued within 24 to 36 months post-implementation at a permanently lower level. Although we do not test for any discontinuity, this can be further explored as more data is released over time.

The number of existing policyholders also dropped, but by less. By October 2024, the number of existing PIF declined by 5% in the Q2 treatment group, 9% in the Q3 treatment group, and 13% in the Q4 treatment group – as compared to the control group (Figure 2B). These estimates imply demand elasticities of -0.63 to -0.15 for the Q2 treatment group, -0.26 to -0.10 for the Q3 treatment group, and greater than -0.14 for the Q4 treatment group. Thus far, the decline in existing policies does not appear to have plateaued, as many of these policies will continue to see year-over-year price increases until they reach their full risk-based rate. These results, as well as the results for new policies, are robust to an alternate specification using a Poisson model (Appendix Figure A1).

Within each treatment quartile, the effects of Risk Rating 2.0 on PIF are larger in zip codes with lower median household income (Figure 3C, Figure 3D). Between May 2024 and October 2024, the number of new PIF in zip codes in the lowest income quartile decreased by an average of 25%, 46%, and 60% across treatment groups, whereas the decline in the highest income quartile was only 7%, 18%, and 32%. Differences between income groups for existing policies are similar but more muted. In the lowest income quartile, existing PIF declined by 9%, 12%, and 17% across treatment groups over the same period, and only by 6%, 8%, and 10% in the highest

quartile. Except for existing policies in zip codes with the Q2 treatment, the differences in effects between SFHA and non-SFHA policies are not statistically significant (Figure 3A, Figure 3B).

Discussion

We find that pricing reforms under Risk Rating 2.0 have led to an 11 – 39% decline in new policies and a 5 – 13% decline in existing policies, depending on how much premiums have increased (Figure 2). These negative effects on NFIP uptake exacerbate previously existing flood insurance protection gaps (Amornsiripantich *et al.*, 2025), and may worsen recovery outcomes following flood events (You and Kousky, 2024). Among existing policies, our estimates correspond to demand elasticities that are closely aligned with the results from Bradt *et al.*, (2021) and Wagner (2022a), and that are generally consistent with other previous studies that find decreases in demand for flood insurance as prices increase (Ortega and Petkov, 2025; Collier *et al.*, 2023; Hennighausen *et al.*, 2023). However, even under Risk Rating 2.0, NFIP demand remains relatively price inelastic, suggesting that low flood insurance uptake is also due to non-price factors, such as informational frictions and cognitive biases (Bradt *et al.*, 2021; Mulder, 2024; Wagner, 2022a).

While the decline in new policies seems to have dropped to a new lower level and has perhaps stabilised, the decline among existing policyholders will likely persist as premiums increase. Given that risk-based pricing is still being phased-in for many existing policies, we expect that the number of NFIP PIF will continue to decrease beyond what has been observed to date. Also, the current risk-based premiums under Risk Rating 2.0 only reflect risk in the present. As flood risk increases under climate change, premiums will likely need to continue to rise, which may in turn decrease policy uptake even further.

As expected, demand for flood insurance is more elastic in lower income zip codes where household budgets are more constrained (Figure 3C; Figure 3D). This aligns with recent research showing that demand elasticities for homeowners insurance coverage vary based on household income (Sastry *et al.*, 2025). More broadly, our re-

sults are consistent with models of insurance demand that emphasise the role of financial constraints in household coverage choices in response to rising premiums (Rampini and Viswanathan, 2013; Ericson and Sydnor, 2018). By contrast, our results challenge the validity of traditional models of rational insurance demand, which assume that households' willingness-to-pay will increase in response to rising risks, and where risk-averse households can frictionlessly borrow more to finance rising premiums (Arrow, 1963; Mossin, 1968; Einav *et al.*, 2010).

We do not find statistically significant differences in the effects of Risk Rating 2.0 between SFHA and non-SFHA properties. This may indicate inconsistent compliance with the NFIP's mandatory purchase requirement (Amornsiripanitch *et al.*, 2025; Blicke and Santos, 2022; Government Accountability Office, 2021), or a growing proportion of SFHA homeowners without federally-backed mortgages – both of which may allow households to drop coverage in response to rising premiums. These results also raise questions about the salience of flood risk information provided by a property's SFHA status in household decisions about purchasing flood insurance (Pollack *et al.*, 2023).

The decline in NFIP uptake, particularly within the SFHA, could be partially, but not fully, explained by households switching to the private insurance market. Between December 2021 and December 2023, the total number of residential NFIP PIF decreased from 4.553 million to 4.329 million. Over that same period, the National Association of Insurance Commissioners (NAIC) private flood insurance data call indicates that the number of private first-dollar⁴ residential flood insurance PIF increased from 300,000 to 375,000 (NAIC, 2023). While the loss of 224,000 NFIP policies is somewhat offset by the gain of 75,000 private policies, there is still a net decrease of 149,000 PIF. Unfortunately, the available data do not allow us to know whether households are simply switching from the NFIP to private companies, or whether private companies are access-

ing a new, previously uninsured segment of the residential market.

The negative effects of Risk Rating 2.0 on the renewal of existing policies may also be partially explained by household relocation. Following previous NFIP price increases, approximately one-third of policy non-renewals were associated with home sales, which suggests that some households may be adapting to increasing rates by moving to lower-risk areas or more resilient homes (Collier *et al.*, 2023). Ge *et al.*, (2025) similarly find that increasing homeowners insurance rates cause households to relocate to properties with lower premiums. However, households that relocate as an adaptation strategy are often replaced by new homeowners who are less likely to be insured (Collier *et al.*, 2023). This indicates that NFIP rates may only affect coverage levels, but have no bearing on the total population exposed to flood risk, despite possible changes in residential sorting (Bakkensen and Ma, 2020). Households could also hypothetically invest in flood risk reduction measures when they choose not to insure in response to rate increases, but there is no evidence of this occurring (Collier *et al.*, 2023), and prior work has found that households use hazard mitigation and insurance as complements, not substitutes (Laird *et al.*, 2021).

These results indicate trade-offs between risk-based pricing and insurance uptake. While risk-based pricing has the potential to create efficiency gains by incentivising adaptation to flood risk and reducing the costs associated with servicing the NFIP's debt to the U.S. Treasury, the decline in insurance coverage may create a range of negative externalities. Following flood events, lack of insurance coverage has negative impacts on individual household recovery (Collier and Kousky, 2024; Kousky, 2019; You and Kousky, 2024), which may worsen socioeconomic inequality (Rhodes and Besbris, 2022). Insufficient coverage can also have negative spillover effects on the surrounding community and local economy in the wake of disaster (You and Kousky, 2024; Box-Couillard and Xu, 2024), and

4. "First-dollar" refers to policies that cover first dollar losses and that are not contingent on NFIP coverage or other coverages that exceed NFIP policy limits. By contrast, "excess" policies provide coverage for losses above NFIP policy limits.

may place greater strain on federal disaster aid and other social safety net program (Deryugina, 2017). Further, underinsurance can create costs for mortgage lenders, as homeowners without insurance are more likely to prepay their mortgage or become delinquent or default following floods and wildfires (Biswas *et al.*, 2023; Kousky *et al.*, 2020; Mota and Palim, 2024), creating broader financial risks.

To manage these trade-offs and mitigate the adverse effects created by rising premiums and lack of insurance, policymakers could consider a means-tested program to help low-income households maintain their flood insurance coverage. While proposals vary in terms of eligibility and level of benefit, the general principle is that households with income under a certain threshold would receive some level of assistance in paying premiums (FEMA, 2018; National Research Council, 2015; U.S. Government Accountability Office, 2016). Akin to other social safety net programs, this affordability program could be supported through general tax revenue

– not by cross-subsidies within the program, as those would only create further distortions and lower insurance uptake (Kousky and Kunreuther, 2014). There is no research to indicate that helping low-income households with the cost of insurance would create moral hazard, but more study is warranted. Insurance uptake could theoretically also be improved through other regulatory and enforcement mechanisms, but absent protective measures for low-income households may only worsen broader housing affordability issues.

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